

Numerical Methods for Computer Science

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Contents

1	Introduction	3
1.1	Why Numerical Methods in Computer Science	3
1.1.1	Approximation Strategies: Promises and Limitations	3
1.1.2	Core Topics in Numerical Methods	3
1.1.3	Key Design Goals in Numerical Methods	4
1.2	Rounding Errors	4
1.2.1	Machine Numbers	4
1.2.2	Numerical Differentiation	5
1.2.3	Convergence Acceleration via Richardson Extrapolation	6
1.3	Matrices & Compute-Work	7
2	Interpolation	8
2.1	Newton	8
2.2	Lagrange	9

1 Introduction

1.1 Why Numerical Methods in Computer Science

Numerical Analysis (*Numerische Methoden*) focuses on developing efficient algorithms and accurate approximations to solve mathematical problems that cannot be solved analytically.

1.1.1 Approximation Strategies: Promises and Limitations

A central task in numerical methods is function approximation. Depending on the properties of the target function and the domain of interest, different representation bases are chosen:

- **Taylor Polynomial Approximation:** Local polynomial approximations rely on Taylor series expansions. While effective locally, Taylor polynomials require high-order derivatives and strict smoothness assumptions ($f \in C^{m+1}$). Furthermore, convergence is strictly local around an expansion point x_0 .
- **Global Polynomial Spaces & Projection:** To obtain global approximations without requiring derivatives, one can select a finite-dimensional subspace (e.g., polynomials up to degree m) spanned by a chosen basis, such as monomials $\{1, x, x^2, \dots, x^m\}$ or orthogonal polynomials. The best approximation is then found by projecting the function onto this subspace under a chosen norm (e.g., L^2 -norm).
- **Fourier Approximation:** For periodic functions, Fourier series provide a global approximation by expanding $f \in L^2(0, 1)$ in terms of complex exponentials:

$$f(t) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{2\pi i k t}, \quad \text{where } \hat{f}(k) = \int_0^1 f(t) e^{-2\pi i k t} dt$$

Fourier approximations achieve rapid (exponential) convergence for smooth periodic functions without requiring explicit derivative calculations.

1.1.2 Core Topics in Numerical Methods

Numerical analysis encompasses several fundamental domains aimed at constructing scalable and stable algorithms:

- **Interpolation:** Constructing an approximating function that passes exactly through a given set of discrete data points (x_j, y_j) .
- **Numerical Quadrature:** Efficiently evaluating definite integrals $\int_a^b f(t) dt$ using weighted sums at discrete node points.
- **Numerical Linear Algebra:** Solving large linear systems $A\underline{x} = \underline{b}$ efficiently and stably.
- **Linear Least Squares (*Lineare Ausgleichsrechnung*):** Finding optimal parameters by minimizing $\|A\underline{x} - \underline{b}\|_2$ for overdetermined systems.
- **Nonlinear Systems of Equations & Least Squares:** Finding roots $\underline{f}(\underline{x}^*) = \underline{0}$ or solving non-linear optimization problems $\min_{\underline{p}} \|f(\underline{p}, \cdot) - y\|_2$.

1.1.3 Key Design Goals in Numerical Methods

When designing numerical algorithms, the main objectives are:

1. **Efficiency & Time Complexity:** Minimizing computation time and floating-point operations.
2. **Memory Footprint:** Optimizing memory usage for large-scale calculations.
3. **Quantifiable Accuracy & Convergence:** Guaranteeing and bounding errors for convergent sequences.
4. **Generality & Stability:** Ensuring robustness against rounding errors and subtractive cancellation.

1.2 Rounding Errors

Definition 1.1. Absolute Error: $\|\tilde{x} - x\|$ with $\|\cdot\|$ a norm in $\mathbb{R}^n$.

Definition 1.2. Relative Error: $\frac{\|\tilde{x} - x\|}{\|\tilde{x}\|}$ for $\tilde{x} \neq 0$, with $\|\cdot\|$ a norm in $\mathbb{R}^n$.

1.2.1 Machine Numbers

Remark 1.3. In modern computer systems, numbers are represented as floating-point numbers using a mantissa-exponent representation.

Definition 1.4. The **Mantissa-Exponent** representation of a number is $\pm M \cdot B^E$, where M is the mantissa containing the significant digits and precision, $B \geq 2$ is the base (typically 2 or 10), and $E \in \{e_{\min}, \dots, e_{\max}\}$ is the exponent determining the scale or magnitude of the number by shifting the radix point.

Example 1.5. Cancellation in Quadratic Equations: The polynomial $p(x) = (x - c)(x - 1/c) = x^2 - (c + 1/c)x + 1$ has roots c and $1/c$. When using the standard quadratic formula (*Mitternachtsformel*) to calculate these roots: The relative error for the smaller root $1/c$ is significantly larger than for the larger root c . This occurs because calculating the smaller root requires subtracting two nearly equal values ($-a$ and $\sqrt{D}$), leading to a loss of significant digits.

Code Example 1.6

```
1 D = a**2 - 4*b
2 wD = sqrt(D)
3 z = ((-a + wD)/2, (-a - wD)/2)
```

Definition 1.7. This phenomenon is called **cancellation** (*Auslöschung*). It describes how the subtraction of two nearly equal numbers dramatically amplifies relative errors, even though the error of the subtraction operation itself is negligible.

Remark 1.8. Stable Alternative via Vieta's Formulas: To avoid cancellation when solving $x^2 + ax + b = 0$, compute the numerically stable larger root x_2 first, and obtain the smaller root x_1 using $x_1x_2 = b \implies x_1 = b/x_2$:

Code Example 1.9

```
1 x2 = (sqrt(D) - a)/2 if a < 0 else (-sqrt(D) - a)/2
2 x1 = b / x2
```

1.2.2 Numerical Differentiation

Remark 1.10. We can use the forward difference quotient to approximate local derivatives:

$$f'(x) \approx \frac{f(x+h) - f(x)}{h}$$

Code Example 1.11

```
1 f = exp; df = exp; x = 0; h = 0.1
2 for k in range(1, 21):
3     ndf = (f(x+h) - f(x)) / h
4     h *= 0.1
```

$\log_{10}(h)$	relativer Fehler
-1	0.05170918075648
-2	0.00501670841679
-3	0.00050016670838
-4	0.00005000166714
-5	0.00000500000696
-6	0.00000049996218
-7	0.00000004943368
-8	0.00000000607747
-9	0.000000008274037
-10	0.000000008274037
-11	0.000000008274037
-12	0.00008890058234
-13	0.00079927783736
-14	0.00079927783736
-15	0.11022302462516
-16	1.00000000000000

Remark 1.12. Observe that the approximation error initially decreases as h grows smaller, reaching optimal accuracy around $h \approx \sqrt{2 \cdot \text{eps}} \approx 10^{-8}$. However, as $h \rightarrow 0$ further, cancellation errors dominate because $f(x+h) \approx f(x)$, causing total accuracy to rapidly degrade.

Theorem 1.13. Complex Step Derivative: If f is analytic, we can use a complex step to perform numerically stable differentiation without cancellation:

$$f(x_0 + ih) = f(x_0) + ihf'(x_0) - \frac{h^2}{2}f''(x_0) - \frac{ih^3}{6}f'''(x_0) + \mathcal{O}(h^4)$$

Taking the imaginary part yields:

$$\begin{aligned} \text{Im}(f(x_0 + ih)) &= hf'(x_0) - \frac{h^3}{6}f'''(x_0) + \mathcal{O}(h^5) \\ \implies f'(x_0) &= \frac{\text{Im}(f(x_0 + ih))}{h} + \mathcal{O}(h^2) \approx \frac{\text{Im}(f(x_0 + ih))}{h} \end{aligned}$$

Since this formulation involves no subtraction of function values, it is immune to subtractive cancellation for arbitrarily small values of h .

1.2.3 Convergence Acceleration via Richardson Extrapolation

Theorem 1.14. Symmetric Difference Quotient and Error Expansion: Using the central difference quotient $d(h) = \frac{f(x+h) - f(x-h)}{2h}$, Taylor expansion yields an error expansion containing only even powers of h :

$$f'(x) - d(h) = c_1h^2 + c_2h^4 + c_3h^6 + \dots + \mathcal{O}(h^{2n+2})$$

Theorem 1.15. Richardson Extrapolation: By combining approximations evaluated at step sizes h and $h/2$, the leading error term c_1h^2 can be eliminated:

$$d_1(h) = \frac{4d(h/2) - d(h)}{3}$$

The refined approximation $d_1(h)$ achieves an increased convergence order of $\mathcal{O}(h^4)$:

$$f'(x) - d_1(h) = \tilde{c}_2h^4 + \tilde{c}_3h^6 + \dots$$

Definition 1.16. Richardson Scheme: Higher-order accuracy can be computed recursively using the triangular scheme ($R_{l,k}$ table) for levels $l = 0, 1, 2, \dots$ and extrapolation steps $k = 1, 2, \dots$:

$$R_{l,0} = d\left(\frac{h}{2^l}\right), \quad R_{l,k} = \frac{4^k R_{l,k-1} - R_{l-1,k-1}}{4^k - 1}$$

Each column k increases the order of convergence such that:

$$f'(x) - R_{l,k} = \mathcal{O}\left(\left(\frac{h}{2^l}\right)^{2k+2}\right)$$

This allows reaching high numerical precision at significantly larger step sizes h , effectively suppressing rounding and cancellation errors.

1.3 Matrices & Compute-Work

Definition 1.17. In NumCS we define the Compute-Work as the theoretical, asymptotic number of elementary operations.

Remark 1.18. Observe that this definition has (almost) nothing to do with the actual compute-time on a chip.

Example 1.19. Worst case Compute-Work for Gauss-Elimination

- General: $O(n^3)$
- Upper Triangular: $O(n^2)$
- Diagonal: $O(n)$

Remark 1.20. Numpy does not automatically adjust algorithms to the given inputs. We have to manually exploit the structure of ex. matrices and vectors.

Example 1.21. Let D be a diagonal matrix and let A be some other matrix. Using `np.dot(D, A)` will result in $O(n^3)$ runtime just like any normal matrix multiplication. We have to manually exploit the structure of D by broadcasting the diagonal entries to a vector and then use numpy to multiply this vector by each row.

Example 1.22. Let $R = \text{triu}(AB)$ be the uppertriangular matrix of the matrix product of $A, B \in \mathbb{R}^{p,n}$ calculating $y = Rx$ takes $O(pn^2)$ using a standard numpy implementation

$$y = \text{np.triu}(A @ B.T) @ x$$

Let $p = 1$ we can represent rewrite this to $A' @ T @ B' @ x$ where A', B' are diagonal matrices made from the rank one matrices A, B , T is a uppertriangular matrix filled with ones. Observe that these products can be just (reverse) partialsums we can calculate using cumulative sum in numpy in $O(n)$. We can use this concept for any p and get a runtime of $O(pn)$.

2 Interpolation

Definition 2.1. Interpolation: Given exact values

$$y_1 = f(x_1, x_2 \dots), y_2 = f(x_i, x_{i+1} \dots) \dots$$

of an unknown function f determine an interpolant $\tilde{f}$ such that

$$\tilde{f}(x_1, x_2, \dots) = y_1$$

Remark 2.2. We assume f to have specific properties (smooth, monotone, etc.) $f \in V$ and choose a Subspace $\tilde{f} \in V_n$ together with a basis $b \in V_n$

Example 2.3. If we choose V_n as the set of all polynomials of degree $\leq n$ we are calculating a polynomial interpolation of f .

Remark 2.4. Different spaces and basis are possible and used for different applications.

Example 2.5. Using $n + 1$ datapoints we can solve the linear system of equations of the $n + 1$ datapoints to get a polynomial of degree n .

$$\begin{aligned} a_n x_1^n + a_{n-1} x_1^{n-1} \dots a_0 &= y_1 \\ a_n x_2^n + a_{n-1} x_2^{n-1} \dots a_0 &= y_2 \\ &\dots \\ a_n x_{n+1}^n + a_{n-1} x_{n+1}^{n-1} \dots a_0 &= y_{n+1} \end{aligned}$$

Definition 2.6. The matrix A formed by this system of linear equations is called a **Vandermonde matrix**.

Remark 2.7. Solving this takes $O(n^3)$ and might come with unreasonably large rounding error.

2.1 Newton

Definition 2.8. Newton-Basis: The newton basis $N_n(x)$ is

$$\prod_{i=0}^{n-1} (x - x_i)$$

Theorem 2.9. Using the newton basis we can continuously increase the degree of interpolation without having to solve the whole system of equations at once.

$$p(x_0) = y_0 \implies p_0(x) = y_0$$

$$p(x_0, x_1) = p_0, y_1 \implies p_1(x) = p_0(x) + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

$$p(x_0, x_1, x_2) = p_1, y_2 \implies p_2(x) = p_1(x) + \frac{\frac{y_2 - y_1}{x_2 - x_1} - \frac{y_1 - y_0}{x_1 - x_0}}{x_2 - x_0} (x - x_0)(x - x_1)$$

...

Remark 2.10. This can be interpreted as

$$p_i(x) = b_1 + b_2 N_1 + b_3 N_2 + \dots$$

where b_i is the i-th divided difference defined by the following recurrence relation

$$b_i = f[x_0, \dots, x_i] = \frac{f[x_1, \dots, x_i] - f[x_0, \dots, x_{i-1}]}{x_i - x_0}$$

and $f[x_i] = y_i$

2.2 Lagrange

Definition 2.11. Lagrange-polynomials are polynomials of the form

$$l_i(x) = \prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j}$$

Theorem 2.12. Observe that for such a Lagrange polynomial $l_i(x)$ we have

$$l_i(x) = 0 \forall x_j \text{ with } j \neq i$$

and

$$l_i(x_i) = 1$$

Theorem 2.13. Because the because the above holds, given a set of datapoints $(x_1, y_1) \dots (x_n, y_n)$ we can construct a polynomial that fullfills all of the n conditions

$$p_n(x) = \sum_{i=0}^n y_i l_i(x)$$